LAB WORKSHEET

Packet Analyser Lab

Capture and analyse traffic from a pretend office network in a Wireshark-style packet analyser.

What you will learn

- How a capture works: the analyst laptop sits on the switch's **mirror (SPAN) port** and sees a copy of every frame.
- The three panes: **packet list**, **packet details** (one layer per line: Frame → Ethernet → IP → TCP/UDP → application), and the **packet explanation**.
- Capture filters (before capturing) vs display filters (after).

Display filters

```
dns          http          tls          arp          dhcp          icmp
ip.addr == 192.168.1.21        ip.src == 192.168.1.22 && dns
tcp.port == 80                 tcp.port in {80 443}
tcp.flags.syn == 1 && tcp.flags.ack == 0     (first packet of each handshake)
tcp.flags.reset == 1           http.request.method == "GET"
http.response.code == 404      dns.qry.name contains "example"
!(arp || dns)                  tcp.stream eq 0
```

Green bar = valid filter; red = invalid (the status bar says why). Use **==**, not =.

Guided steps

1. Start a capture on eth0.
2. Choose HTTP + HTTPS in the traffic generator and generate.
3. Stop the capture.
4. Filter `dns`; find the IP of intranet.oakfield.test.
5. Filter SYN-only packets; find the server port.
6. Filter `http`, expand the HTTP layer.
7. Follow the TCP stream.
8. Follow the TLS stream — can you read it?
9. Statistics → Protocol Hierarchy.
10. Filter with `ip.addr`.
11. Save / export the capture.

Independent task

Open **helpdesk-tuesday.cap** and solve the tickets: a laptop with no DHCP lease, a PC using the wrong DNS server, a printer that never answers ARP, a 404 page, a refused TCP port and a site still using unencrypted HTTP.